

SUNDSVALL RASTA V, Sweden

WMO: 023540

Lat: 62.4306N

Lon: 17.3447E

Elev: 262

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.0	2.2	-6.7	3.9	-1.8	-2.3	4.9	2.9	16.3	29.7	14.5	29.0	3.0	0	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	78.0	63.9	74.6	62.4	71.5	60.6	66.0	73.9	64.0	71.8	62.3	69.0	5.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	86.7	69.4	61.1	81.3	67.4	59.3	76.0	65.4	30.8	73.7	29.4	72.1	28.1	69.1	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.2	12.2	10.9	DB	-5.8	83.9	6.7	3.2	-10.6	86.3	-14.5	88.1	-18.3	89.9	-23.2	92.3	(4)
			WB	-5.6	68.3	6.6	2.0	-10.4	69.7	-14.3	70.9	-18.0	72.0	-22.8	73.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	22.6	23.8	31.1	38.8	47.9	55.9	62.1	59.4	51.3	40.8	32.5	26.0
	DBStd	15.33	9.94	10.11	7.69	5.93	6.15	5.42	4.85	4.80	5.61	6.42	7.52	10.00
	HDD50	4240	848	732	585	341	115	9	0	1	51	288	525	745
	HDD65	8759	1313	1152	1050	787	530	278	120	185	411	749	974	1210
	CDD50	999	0	0	0	4	51	184	374	292	90	4	0	0
	CDD65	42	0	0	0	0	0	4	28	10	0	0	0	0
	CDH74	355	0	0	0	0	10	74	216	55	0	0	0	0
	CDH80	51	0	0	0	0	0	11	35	5	0	0	0	0
Wind	WSAvg	4.8	4.9	4.6	5.2	5.0	5.1	5.4	4.8	4.4	4.6	4.6	4.8	4.6
Precipitation	PrecAvg	29.4	2.1	1.6	1.8	1.9	1.7	2.2	3.0	3.4	3.1	3.0	3.2	2.4
	PrecMax	39.6	5.6	4.7	4.9	4.9	4.9	6.1	8.0	10.0	10.2	8.0	8.7	9.3
	PrecMin	18.7	0.0	0.2	0.3	0.2	0.2	0.2	0.3	0.6	0.3	0.5	0.7	0.2
	PrecStd	4.9	1.3	1.0	1.1	1.3	0.9	1.4	1.8	2.2	1.9	1.9	2.0	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	45.6	53.7	65.1	74.7	80.9	83.9	79.6	70.6	58.6	51.3	46.2
		MCWB	38.7	40.0	45.1	52.5	60.2	64.0	66.1	62.8	59.3	51.1	47.0	42.2
	2%	DB	39.9	42.1	48.8	58.9	67.7	75.3	79.6	75.1	66.0	55.4	48.0	42.9
		MCWB	36.7	37.7	41.3	48.4	55.5	61.2	64.9	62.8	57.2	49.8	44.5	39.4
	5%	DB	37.6	39.2	45.1	53.9	63.6	71.1	75.9	71.8	63.0	52.5	44.4	40.7
		MCWB	34.8	35.6	39.1	45.1	52.5	58.4	63.6	61.5	55.5	48.8	41.9	37.8
	10%	DB	35.1	36.5	42.2	50.6	59.9	67.3	72.7	68.6	60.2	50.3	41.9	38.2
		MCWB	33.0	33.6	37.1	42.4	50.4	56.4	62.1	60.2	54.2	47.1	40.0	35.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.3	40.5	45.3	53.0	61.0	65.2	69.2	68.2	60.8	53.8	48.0	42.4
		MCDB	42.0	45.0	53.3	62.8	72.4	77.9	79.9	75.4	68.4	56.9	50.7	45.7
	2%	WB	37.2	38.3	41.8	49.2	56.3	62.2	66.6	65.1	58.5	51.0	44.9	40.0
		MCDB	39.6	41.6	48.0	57.8	66.0	72.9	75.2	70.6	63.5	53.5	47.5	42.4
	5%	WB	35.3	36.0	39.5	45.8	53.5	59.8	65.1	63.0	56.9	49.4	42.4	38.1
		MCDB	37.2	38.8	44.5	52.9	62.2	68.8	73.6	68.3	61.0	52.3	44.0	40.4
	10%	WB	33.1	33.8	37.4	43.1	51.0	57.6	63.4	61.4	55.4	47.3	40.4	36.0
		MCDB	34.7	36.2	41.9	49.2	58.7	65.8	71.0	66.8	59.1	50.1	41.7	37.9
Mean Daily Temperature Range	5% DB	MDBR	9.3	10.8	13.0	15.2	16.3	16.5	15.7	14.6	13.8	11.1	7.5	9.0
		MCDBR	8.8	11.6	16.3	23.0	23.1	23.9	20.8	19.2	17.2	14.4	10.0	11.1
		MCWBR	7.6	9.1	11.0	13.6	12.5	11.8	9.9	9.3	10.0	9.9	8.1	9.4
	5% WB	MCDBR	8.5	10.9	15.9	22.1	21.7	21.9	18.7	15.6	14.9	13.5	9.4	10.6
MCWBR		7.5	8.8	11.0	13.8	12.6	11.5	9.9	8.7	9.8	10.0	8.1	9.2	
Clear-Sky Solar Irradiance	taub		0.237	0.293	0.316	0.352	0.354	0.349	0.366	0.358	0.325	0.311	0.254	0.345
	taud		2.113	2.252	2.351	2.378	2.437	2.485	2.456	2.473	2.548	2.449	2.180	2.624
	Ebn at Noon		139	210	252	265	274	278	269	261	251	206	130	86
	Edh at Noon		10	19	25	30	31	30	31	28	21	16	10	6
All-Sky Solar Radiation	RadAvg		64	268	736	1195	1523	1759	1609	1222	769	337	95	35
	RadStd		9	43	82	102	157	111	173	102	95	60	19	3

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page